

Biosketch — Nawder Loswin

TriadicFrameworks Research Initiative

ORCID: 0009-0002-2282-5460

Professional Overview

Nawder Loswin is the founder and lead architect of the TriadicFrameworks Research Initiative, an open-science effort focused on developing substrate-agnostic frameworks for regime-aware sensing, modeling, and simulation. His work centers on creating minimal, interoperable scientific protocols that support transparent, reproducible analysis across remote sensing, planetary science, and autonomous exploration systems.

Loswin's contributions include the design and publication of the Dimensional Substrate Regime Scanning Protocol (dsrsp/0.1), the Resonance Substrate Model (RSM), the Validation-Space-Time engine (vST), and the Structural Life-Regime Profiles (SLRP). These components form a coherent open-science substrate that enables consistent interpretation of structural, sensory, and environmental patterns across scientific domains.

Research Interests

- open-science infrastructure and reproducible scientific workflows
- regime-aware sensing and structural analysis
- substrate-agnostic scientific protocols
- simulation-based validation and training environments
- planetary regime modeling and autonomous exploration systems

Relevant Experience

Founder and Lead Architect, TriadicFrameworks Research Initiative

- Designed and published dsrsp/0.1, a minimal wire format for regime-aware sensor interpretation.
- Developed the RSM structural grammar for coherence, drift, and stability analysis across physical and simulated systems.
- Created the vST validation framework for analyzing regime transitions and dimensional continuity.
- Authored the Structural Life-Regime Profiles (SLRP) for biological, synthetic, and planetary systems.
- Published more than twenty open-science artifacts with DOIs through Zenodo, establishing a coherent and extensible research canon.
- Built documentation, schemas, integration guides, and onboarding materials to support community adoption and reproducibility.

Selected Open-Science Outputs (DOIs)

A complete list is provided in `references/triadicframeworks_doi_list.md`.

Representative examples include:

- Loswin, N. (2026). *Dimensional Substrate Regime Scanning Protocol (dsrsp/0.1)*. Zenodo. DOI: 10.5281/zenodo.18331997
- Loswin, N. (2025–2026). TriadicFrameworks open-science canon (multiple releases). Zenodo. Various DOIs.

Skills and Expertise

- protocol and schema design
- open-source software development
- scientific documentation and specification writing
- structural and regime-based analysis
- simulation integration and workflow design
- community-oriented open-science practices

Contribution to the Proposed Project

As Principal Investigator, Loswin will lead the scientific and technical development of all TriadicFrameworks components, oversee documentation and integration efforts, coordinate open-source releases, and ensure that all deliverables align with NASA's open-science mission. His experience designing substrate-agnostic frameworks and publishing open scientific artifacts provides a strong foundation for successful execution of the proposed work.